Data 102 Project

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1 Data Overview

1.1 Dataset 1: CDC Chronic Disease Indicators (CDI)

The CDC Chronic Disease Indicators dataset provides annual estimates of asthma prevalence at the county level in the United States. For this project, we focused on 2018 California, with the variable Data_Value, which reports the age-adjusted percentage of adults diagnosed with asthma. These estimates originate from the Behavioral Risk Factor Surveillance System (BRFSS), a national telephone survey administered by the CDC. The data are sample-based and modeled to produce reliable small area estimates. After preprocessing, county-level prevalence values were accurately merged with other datasets at the Census Block Group (CBG) level. The CDI data offer essential information on the spatial distribution of chronic respiratory health outcomes. Although the original estimates are subject to sampling error and smoothing bias, careful spatial and temporal alignment ensures their relevance at the finer CBG granularity used in our analysis. Dataset 1

1.2 Dataset 2: CDC PM2.5 Concentrations

The PM2.5 dataset contains model-based predictions of average annual concentrations of fine particulate matter (PM2.5) at the Census Block Group level in California. These estimates are derived from a combination of satellite imagery, chemical transport models, and ground monitor calibration. The DS_PM_pred variable captures long-term exposure to ambient air pollution, a key environmental risk factor in respiratory health. As this dataset is produced using high resolution spatial modeling, it provides a continuous surface of pollution values across all geographic units, including rural and low-monitoring areas. Since the asthma prevalence and demographic data were aligned to the same geographic and temporal resolution, the PM2.5 predictions offer a reliable measure of environmental exposure for causal inference and prediction tasks. While inherent modeling uncertainties exist, they are minimized by the fine spatial granularity and ensemble modeling techniques used in the data source. Dataset 2

1.3 Dataset 3: U.S. Census Demographics (ACS 5-Year Estimates)

The demographic and socioeconomic features used in this study were drawn from the American Community Survey (ACS) 5-Year Estimates, which provide robust data at the Census Block Group level. Key variables include racial composition (White, Black, Hispanic, Asian, Native, Pacific), economic indicators (IncomePerCap, ChildPoverty), and occupational structure (Professional, Production). These variables were selected for their relevance to health disparities and environmental

justice. The ACS data are based on large-scale probability samples and are modeled to yield accurate estimates even for small geographic units like CBGs. By using 5-year estimates, we ensured greater stability and reliability in low-population or rural areas. Though ACS data are subject to sampling error and margins of uncertainty, these were minimized in our analysis through appropriate aggregation and temporal matching. Overall, the ACS provides critical context for understanding how social determinants intersect with environmental exposures and health outcomes. Dataset 3

1.4 Objective

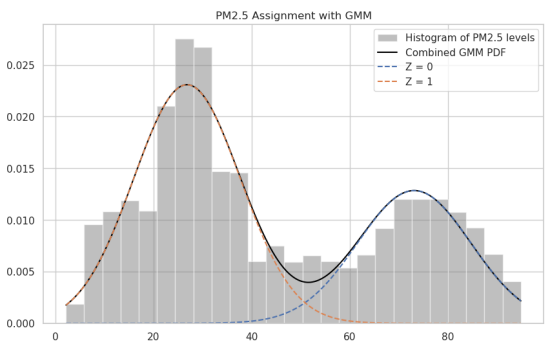
The primary goal of our project is to investigate the relationship between air pollution, specifically long-term PM2.5 exposure and asthma prevalence in California, with a dual focus on causal inference and predictive modeling. We aim to estimate the causal effect of high PM2.5 levels on asthma prevalence, controlling for demographic and socioeconomic confounders, and to assess how well asthma or pollution levels can be predicted using features such as race, income, and occupation. To enable this analysis, we merged three datasets at the Census Block Group (CBG) level: dataset 1 modeled asthma prevalence data from the CDC Chronic Disease Indicators (CDI), dataset 2 predicted annual average PM2.5 concentrations from satellite-based models, and dataset 3 demographic and economic data from the American Community Survey (ACS) 5-Year Estimates. The merge was performed using the 10-digit GEOID, a unique identifier assigned to each CBG that ensures geographic consistency across datasets. All data were temporally aligned to reflect the most recent overlapping years available. Because asthma data were originally reported at the county level, we uniformly assigned those prevalence rates to all CBGs within a given county, enabling analysis at a finer resolution while preserving regional trends. This harmonization process allowed us to construct a unified, high-resolution panel dataset suitable for both statistical estimation and modeling approaches.

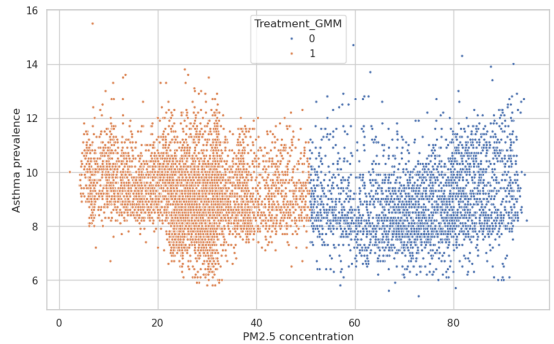
1.5 Research Questions

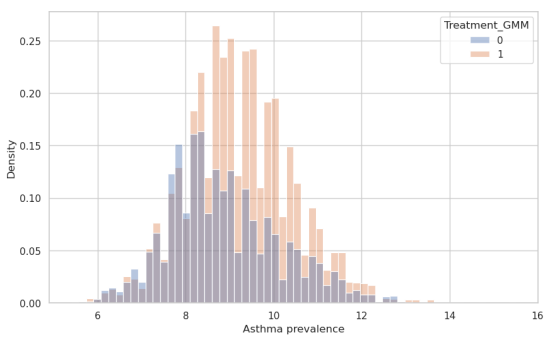
1. **Causal Inference:** In 2018, what is the causal effect of high predicted PM2.5 levels on adult asthma prevalence in California?
2. **Prediction with GLMs and nonparametric methods:** Can we predict air pollution levels (PM2.5) using asthma prevalence and socioeconomic/demographic indicators at the Census Block Group level in California (2018)?

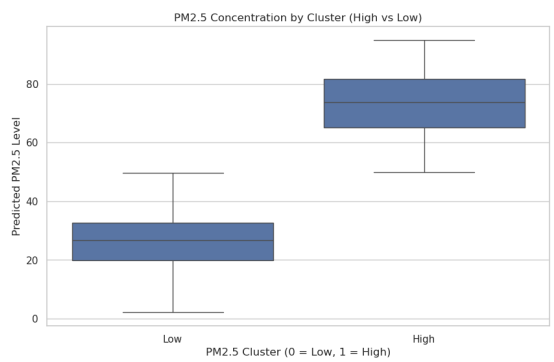
2 EDA

2.1 Binarization of PM2.5 levels (Causal Inference): Clustering and Distribution





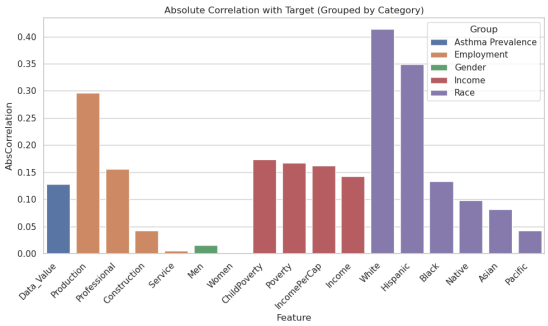


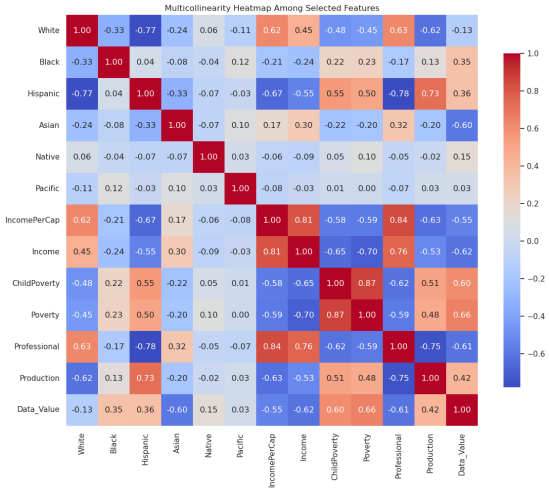


- A Gaussian Mixture Model (GMM) was used to fit two distributions to the bimodal distribution of PM2.5 concentrations, then those two distributions were used to assign data points to treatment groups. Each data point was assigned to the group whose distribution had a higher PDF at that point. The distribution with the lowest mean (left peak in the graph above) corresponds to low PM2.5 concentrations, and therefore the group where treatment equals 0 since the treatment is being in a high PM2.5 area.
- The plot clearly confirms that the PM2.5 clustering (into "High" and "Low") is meaningful and distinct. The median PM2.5 level in the high cluster is substantially higher (around 75) than in the low cluster (around 28). There is little overlap in interquartile ranges (IQRs), which suggests strong separation between the two groups.
- This clustering provides a binary treatment variable (High vs. Low PM2.5) necessary for methods like Inverse Propensity Weighting or matching. It validates that the treatment is not arbitrary— we're distinguishing areas that truly differ in pollution exposure. This setup helps us estimate the causal effect of high PM2.5 on asthma while adjusting for confounders

like race or gender.

2.2 Feature Selections (Prediction with GLMs and nonparametric methods)





Step 1: Absolute Correlation Analysis

We began by calculating the absolute correlation between each candidate feature and the target variable, PM2.5, to understand which features are most linearly associated with pollution levels.

- We plotted a grouped bar chart that visualized absolute correlations, color-coded by feature category (Employment, Gender, Income, Race).
- Features with an absolute correlation above 0.15 with PM2.5 were prioritized for selection due to their stronger individual predictive potential. High-correlation features included: 1.Employment: Production, Professional 2. Income: ChildPoverty, IncomePerCap 3.Race: White, Hispanic

Step 2: Addressing Multicollinearity

Next, we used a correlation heatmap to evaluate multicollinearity among the high-correlation features. This step was essential to avoid instability in regression models and ensure that features provide unique, non-redundant information.

- Strong collinearity was observed among the income-related features: Income, IncomePerCap, Poverty, ChildPoverty.
- To address this, we excluded Income and Poverty. We retained IncomePerCap and ChildPoverty, which had higher absolute correlations with PM2.5 and captured both per-capita and child-focused poverty metrics.

Step 3: Demographic Fairness Considerations

While some race-related variables (Pacific, Asian) had low individual correlations with PM2.5, we intentionally retained all race variables:

- White, Black, Hispanic, Asian, Native, Pacific
- This decision was made to avoid omitting underrepresented populations and to capture any racial disparities in pollution exposure.
- Including all race categories aligns with equity-focused modeling and ensures broader demographic representation.

3 Research Question 1: Causal Inference

3.1 Question

In 2018, what is the causal effect of high predicted PM2.5 levels on adult asthma prevalence in California?

3.2 Introduction

This report investigates the relationship between air pollution and asthma prevalence in California, focusing on causal inference using observational data from 2018. The project aims to estimate the average treatment effect (ATE) of high predicted $PM_{2.5}$ exposure on adult asthma prevalence, while accounting for potential confounding socioeconomic and demographic variables.

Research Question:
Does exposure to high levels of $PM_{2.5}$ (daily average $> 12\ \mu g/m^3$) cause an increase in asthma prevalence among California census tracts in 2018?

Treatment and Outcome

- **Treatment Variable:** PM25_Treatment
  A binary indicator derived from predicted $PM_{2.5}$ values (variable `DS_PM_pred`), binarized into "High" (1) and "Low" (0) exposure categories.
- **Outcome Variable:** Data_Value
  Asthma prevalence rate (percent), obtained from the CDC's chronic disease indicators dataset.

Confounding Variables and Colliders

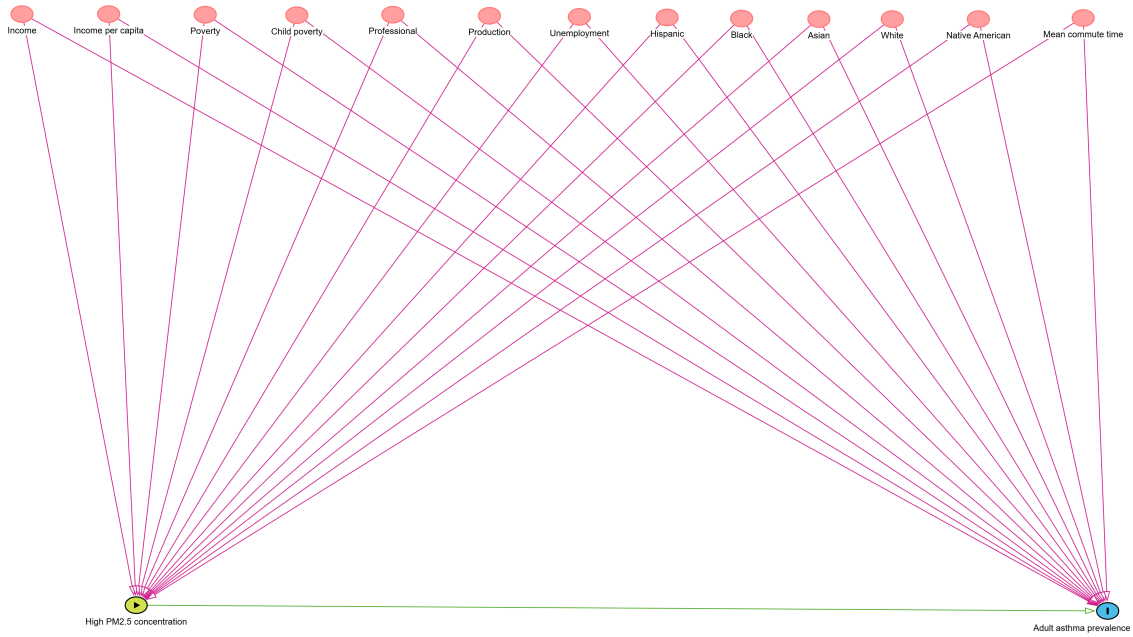
To control for confounding, the following demographic and socioeconomic variables are included:

- **Race:** White, Black, Hispanic, Asian, Native, Pacific
- **Gender:** Men, Women
- **Income:** Income, Poverty, IncomePerCap, ChildPoverty
- **Socioeconomic Controls:** Employment rate, Mean commute time, and job categories such as Professional, Service, Construction, Office, Production

No colliders were found among the variables involved in the experiment performed for this research question.

3.3 Causal Graph

The directed acyclic graph (DAG) below illustrates the causal relationship between the treatment, outcome, and confounding variables in this experiment. An arrow from one variable into another signifies that the former has a causal effect on the latter.



Estimation Method

We use **Inverse Propensity Weighting (IPW)** or **matching methods** to estimate the *Average Treatment Effect (ATE)* of high pollution exposure on asthma rates. These methods adjust for differences in confounders between treated (high pollution) and control (low pollution) groups to approximate the effect of treatment as if it were randomized.

Why Causal Inference?

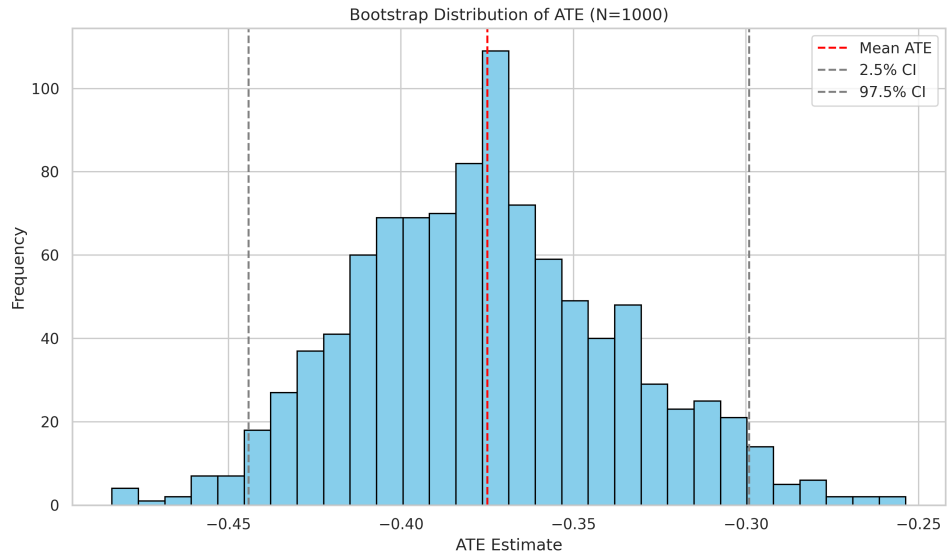
A causal inference approach is appropriate because we are interested in estimating the effect of a modifiable exposure (air pollution) on a health outcome (asthma). While randomized experiments are not feasible in this context, observational data with proper adjustment allows for causal conclusions under assumptions of no unmeasured confounding and correct model specification.

Limitations

- Measurement error in pollution estimates or asthma prevalence could bias the results.
- Unmeasured confounders such as housing quality or healthcare access may violate the ignorability assumption.
- Treatment assignment (high vs. low $PM_{2.5}$) is based on predicted rather than observed concentrations.

3.4 Conclusion

To assess the uncertainty of our estimated average treatment effect (ATE), we performed bootstrap resampling with 1,000 iterations. The histogram below shows the distribution of ATE values generated across these resamples. As shown, the sampling distribution is roughly symmetric and centered near the observed ATE of -0.37. The dashed lines indicate the 2.5% and 97.5% quantiles, forming a 95% confidence interval from -0.44 to -0.30. This visualization supports the robustness and precision of our causal estimate.



Key Findings

Using inverse propensity weighting (IPW), we estimated the average treatment effect (ATE) of high $PM_{2.5}$ exposure on adult asthma prevalence across California census tracts. After adjusting for confounders including income, race/ethnicity, poverty, employment type, and commute patterns, we found an estimated ATE of approximately **-0.38 percentage points**. A bootstrap procedure with 1,000 resamples yielded a 95% confidence interval of **[-0.45, -0.30]**, suggesting the effect is statistically distinguishable from zero.

Surprisingly, the estimated effect was negative: tracts with higher $PM_{2.5}$ levels were associated with slightly lower asthma prevalence. This contradicts prior expectations and public health literature linking air pollution to increased respiratory risk.

Interpretation and Uncertainty

Several explanations are possible for this counterintuitive result. First, the clustering-based definition of high/low $PM_{2.5}$ exposure may not fully capture clinically relevant pollution thresholds. Second, unmeasured confounding (e.g., housing quality, access to healthcare, or differences in population vulnerability) could bias the estimate. Third, structural factors like urban density may

drive inverse associations — urban tracts with high pollution may also have better healthcare infrastructure, mitigating asthma risk.

Despite this, our bootstrapped confidence interval suggests the result is robust under repeated resampling. The DAG we constructed supports our covariate adjustment strategy, and our propensity score model did not condition on colliders.

Modeling Limitations and Assumptions

Our causal claims rely on the **unconfoundedness assumption**, which posits no unmeasured confounding conditional on observed covariates. It also assumes correct specification of the propensity score model. While we took care to avoid colliders and relied on domain knowledge and DAGs to guide our selection, these assumptions are not testable. Our analysis is also limited by its cross-sectional, ecological design and may not generalize to individual-level effects.

Future Work and Implications

This unexpected finding underscores the importance of critical modeling choices in observational inference. Future work should explore alternative causal methods (e.g., matching or doubly robust estimation), test sensitivity to unobserved confounding, and incorporate longitudinal or individual-level data. Replicating this analysis using different definitions of $PM_{2.5}$ exposure or examining subpopulations could help clarify mechanisms.

Ethical and Practical Considerations

From a policy perspective, our results do not provide immediate evidence to deprioritize air pollution interventions, given existing research linking $PM_{2.5}$ to respiratory illness. Instead, they highlight the complexity of causal modeling and the need for caution when interpreting ecological data. We affirm the ethical importance of reducing air pollution as a public good, even as our findings illustrate how localized data may diverge from broader expectations.

4 Research Question 2: Prediction with GLMs and nonparametric methods

4.1 Question

Can we predict air pollution levels (PM2.5) using asthma prevalence and socioeconomic/demographic indicators at the Census Block Group level in California (2018)?

4.2 GLM

4.2.1 Modeling Choices and Implementation

For this analysis, we use a Generalized Linear Model (GLM) with a Gaussian family and identity link function to predict DS_PM_pred (predicted PM2.5 pollution) using a set of demographic and socioeconomic covariates. After comparing two modeling strategies—Plain GLM and Log-transformed + Standardized (Log+Z) GLM—we selected the Log+Z GLM as our final model.

Why a Gaussian GLM?

- Continuous outcome: The response variable, DS_PM_pred, is a continuous prediction of fine particulate matter (PM2.5), making Gaussian an appropriate distribution family.
- Additive relationships: Gaussian GLMs assume linear additive effects of predictors on the outcome, which aligns with our use of standardized demographic features (percent White, percent Professional, per capita income).
- Interpretability: Coefficients in a Gaussian GLM with identity link are straightforward to interpret (change in outcome per unit change in predictor), which aids in policy-relevant interpretation.

Why Log-Transform and Standardize (Log+Z)?

Although the Plain GLM slightly outperformed in RMSE and R^2, the Log+Z GLM was chosen due to substantially better model fit, evidenced by a drastically lower Akaike Information Criterion (AIC).

- AIC Interpretation: AIC balances model fit with complexity; lower values indicate a better fit. The around 47,000-point drop in AIC for the Log+Z model indicates overwhelming support in favor of it, despite marginally worse RMSE.
- Log-transformation: Applying log1p() to DS_PM_pred stabilizes variance and handles right-skewness common in pollution metrics.
- Standardization (Z-score): Standardizing features ensures that all predictors are on the same scale, which improves convergence and interpretability when features vary widely in magnitude (per capita income vs. percentage racial group).

4.2.2 Assumption

- Linearity in parameters: Assumes a linear relationship between the predictors and the log-transformed outcome.

- Homoscedasticity (constant variance): After log-transformation, we assume constant variance in residuals. This was visibly improved post-transformation.
- Independence of errors: Residuals are assumed to be independent across observations. While spatial autocorrelation is a concern in geographic data, this was not explicitly modeled here.
- Normality of errors: GLMs do not require normally distributed residuals, but for inference (z-scores, confidence intervals), approximate normality is assumed—made more plausible with log transformation.
- No multicollinearity: Standardizing predictors helps reduce multicollinearity, but the presence of highly correlated features (income and professional occupation) is still possible and should be monitored.

4.2.3 Performance Part I: Raw

```
=== [PLAIN GLM SUMMARY] ===

                 Generalized Linear Model Regression Results
==============================================================================
Dep. Variable:            DS_PM_pred   No. Observations:                 6403
Model:                           GLM   Df Residuals:                     6392
Model Family:               Gaussian   Df Model:                           10
Link Function:              Identity   Scale:                          475.13
Method:                         IRLS   Log-Likelihood:                -28813.
Date:               Fri, 02 May 2025   Deviance:                   3.0371e+06
Time:                       03:39:22   Pearson chi2:                 3.04e+06
No. Iterations:                    3   Pseudo R-squ. (CS):             0.2735
Covariance Type:           nonrobust
===================================================================================
                   coef    std err          z      P>|z|      [0.025      0.975]
-----------------------------------------------------------------------------------
const          -44.7247     11.044     -4.050      0.000     -66.370     -23.080
White            0.3333      0.115      2.891      0.004       0.107       0.559
Black            0.8743      0.123      7.123      0.000       0.634       1.115
Hispanic         0.8633      0.111      7.752      0.000       0.645       1.082
Asian            0.6617      0.116      5.720      0.000       0.435       0.888
Native           0.3252      0.180      1.806      0.071      -0.028       0.678
Pacific         -1.0442      0.320     -3.262      0.001      -1.672      -0.417
IncomePerCap  2.323e-05   2.63e-05      0.884      0.377   -2.83e-05    7.47e-05
ChildPoverty     0.0293      0.021      1.388      0.165      -0.012       0.071
Professional     0.5620      0.037     15.147      0.000       0.489       0.635
Production       0.6478      0.059     11.060      0.000       0.533       0.763
===================================================================================

=== [LOG+STANDARDIZED GLM SUMMARY] ===

                 Generalized Linear Model Regression Results
```

```
==============================================================================
Dep. Variable:             DS_PM_pred   No. Observations:                 6403
Model:                            GLM   Df Residuals:                     6392
Model Family:                Gaussian   Df Model:                           10
Link Function:               Identity   Scale:                          0.30406
Method:                          IRLS   Log-Likelihood:                -5268.4
Date:                Fri, 02 May 2025   Deviance:                       1943.5
Time:                        03:39:22   Pearson chi2:                 1.94e+03
No. Iterations:                     3   Pseudo R-squ. (CS):             0.3139
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          3.6223      0.007    525.582      0.000       3.609       3.636
x1            -0.0796      0.077     -1.037      0.300      -0.230       0.071
x2             0.1026      0.027      3.757      0.000       0.049       0.156
x3             0.3320      0.075      4.453      0.000       0.186       0.478
x4             0.1190      0.045      2.648      0.008       0.031       0.207
x5            -0.0272      0.008     -3.196      0.001      -0.044      -0.011
x6            -0.0232      0.008     -2.973      0.003      -0.038      -0.008
x7             0.0275      0.013      2.041      0.041       0.001       0.054
x8            -0.0029      0.009     -0.321      0.748      -0.021       0.015
x9             0.2989      0.017     17.544      0.000       0.266       0.332
x10            0.1294      0.011     11.479      0.000       0.107       0.152
==============================================================================
Train/Test Performance
Model          Train RMSE    Test RMSE
Plain               21.78        21.82
Log + Z             22.60        22.56

K-Fold Cross-Validation (5 Folds)
Model         Mean RMSE    Std Dev
Plain             21.90       0.25
Log + Z           22.59       0.12

AIC (Model Fit)
Model               AIC
Plain          57647.42
Log + Z        10558.89
```

4.2.4 Performance Part II: Unforseen Data Prediction (CA 2017, WA 2017)

```
=== Log + Z Model Performance Summary ===
Train (CA 2018) | RMSE:  22.60
Test (CA 2018) | RMSE:  22.56
```

```
WA 2017  | RMSE:  15.07
CA 2017  | RMSE:  32.65
```

4.2.5 Performance Summary

Round 1: Model Summary and Interpretation (Training/Test on CA 2018)

The Log+Z model generalizes well within the same state and year. The negligible RMSE difference between train and test indicates:

- Low overfitting
- Stable relationships between demographic features and predicted pollution levels within CA 2018

Round 2: Generalization to New Regions/Years

We evaluated model performance on out-of-sample data (California 2017 and Washington 2017), testing both spatial and temporal generalization.

- The lower RMSE on WA 2017 suggests that the model captures robust, structural associations between demographics and pollution that hold across states.
- The higher RMSE on CA 2017 may reflect temporal drift (fires, regulatory changes, economic shifts).

4.2.6 Uncertainty and Discussion

```
Closest match:
StateAbbr                 CA
CountyName        Stanislaus
White                   42.5
Black                    1.3
Hispanic                49.1
Asian                    3.5
Native                   2.2
Pacific                  0.0
IncomePerCap           30423
ChildPoverty             3.5
Professional            27.4
Production              28.7
Actual                15.604
Prediction         15.543932
Lower Interval     14.753136
Upper Interval     16.374426
AbsError            0.060068
Name: 8844, dtype: object

Farthest mismatch:
```

```
StateAbbr                CA
CountyName         Imperial
White                  14.8
Black                  15.8
Hispanic               65.5
Asian                   1.7
Native                  0.8
Pacific                 0.0
IncomePerCap           6549
ChildPoverty           29.5
Professional           35.6
Production             15.6
Actual                3.592
Prediction        88.069242
Lower Interval    80.264087
Upper Interval    96.624058
AbsError          84.477242
Name: 2553, dtype: object
```

Closest Match – Stanislaus County, CA

- White: 42.5%, Hispanic: 49.1%, Black: 1.3%

  Moderate to high White and Hispanic population mix, with a very low black population. This aligns with the GLM's feature weights: both White and Hispanic shares are positively associated with PM2.5, so their combination gives the model strong predictive clarity.

- Income Per Capita: $30,423, Child Poverty: 3.5%

  High income and low child poverty indicate socioeconomic advantage, which is associated with lower PM2.5 levels. This likely offsets some of the racial risk factors and helps balance the model's prediction.

- Prediction: 15.54, Actual: 15.60, Abs Error: 0.06

  The model performs extremely well here, with an almost perfect prediction.

- Discussion

  Aleatoric (data noise): Low — the features align cleanly, with no conflicting signals.

  Epistemic (knowledge gap): Low — the county's profile is well represented in the training data.

  Misspecification risk: Minimal — the model has seen similar patterns and responds confidently.

Farthest Mismatch – Imperial County, CA

- White: 14.8%, Hispanic: 65.5%, Black: 15.8%

This is a highly underrepresented and complex demographic mix: very low White percentage, high Hispanic and Black percentage. The GLM associates high Hispanic and Black shares with increased PM2.5, which likely drives the overprediction.

- Income Per Capita: $6,549, Child Poverty: 29.5%

  These values signal deep socioeconomic disadvantage, again strongly associated with high predicted PM2.5 in the model. However, the actual PM2.5 is very low, which breaks model expectations.

- Prediction: 88.07, Actual: 3.59, Abs Error: 84.48

  A catastrophic miss.

- Discussion

  Aleatoric: High — conflicting signals (high professional % vs. very low income) make it hard for the model to settle on a clear prediction.

  Epistemic: High — this feature combination (very low income, high Black and Hispanic %, high professional) is rare or unseen in training data.

  Misspecification: Likely — the GLM assumes additivity and linearity, but this case may require interactions (Hispanic $\times$ Professional = lower pollution if jobs are indoors), which the model can't capture.

4.3 Random Forest

4.3.1 Modeling Choices and Implementation

Final validation of feature description

After discussions with the GLM modeler, we have unanimously decided to use the following 10 ACS public variables as **independent variables in both the Random Forest and GLM lines:**

White, Black, Hispanic, Asian, Native, Pacific, IncomePerCap, ChildPoverty, Professional, Production.

This list balances predictive relevance ($|\rho| > 0.15$ for the core variables), control for multicollinearity (retaining IncomePerCap + ChildPoverty for the income category), ethnic fairness (retaining the full six-ethnicity proportion),and ensure that the two models are compared in a fully consistent feature space.

Random Forest Set Up

We chose **Random-Forest Regressor** as the non-parametric predictive model with the following configuration:

- `n_estimators = 600` - sufficient number of trees to reduce variance and stabilize feature importance;
- `max_depth = 20` - to control the complexity of a single tree to prevent overfitting;
- `max_features = 0.5` - only 50% of features are used per cut to increase diversity;
- `bootstrap = True`, `oob_score = True` - use self-sampling and estimate generalization error in real time using out-of-bag (OOB) samples;
- `random_state = 42`, `n_jobs = 2` - results are reproducible and parallel computation saves time.

Reasons for choice

1. **Nonlinear Capability** Random Forests can automatically capture complex interactions and nonlinear relationships without the need to specify the functional form in advance.
2. **Insensitivity to scales and outliers** Socioeconomic variables span a wide range and may contain extreme values, which tree models can naturally cope with.
3. **Interpretability** Out-of-bag error and permutation importance provide intuitive overfitting monitoring and feature contribution.

4.3.2 Assumptions and Metrics

Implicit assumptions and limitations

- Defaults observations to be independently and identically distributed.

- Continuous high base features may be preferred in splits; Permutation importance magnifies their effect.

Assessment indicator

- **R^2 (coefficient of determination)** Measures the proportion of the variance of the target variable explained by the model;
- **RMSE (Root Mean Square Error, $\mu g/m^3$)** Consistent with the original scale to facilitate interpretation of absolute bias.

Assessment Stratification

1. **5-fold cross-validation (CV)**
   - Performed on 80% of the CA-2018 training set, giving the mean ± standard deviation of R^2 vs. RMSE.
   - Purpose: Estimate the model variance and observe whether the hyperparameters are stable.
2. **Leave the test set**
   - Use the remaining 20% of CA-2018 trajectories that did not appear in the training at all.
   - Purpose: to provide unbiased generalization errors for the same year and state.
3. **External validation (CA-2017 vs. WA-2017)**
   - Load `test.xlsx` and compute RMSE for each of the two states.
   - Purpose: To test the robustness of the model to temporal drift (2018→2017) and spatial drift (CA→WA).
4. **Training set RMSE vs. OOB R^2**
   - The training RMSE is used to quickly see the level of fit;
   - OOB R^2 is automatically given by out-of-bag samples and can be compared to CV/Test to determine overfitting.
5. **Permutation Importance Permutation Importance** **Permutation Importance
   - Randomly select $\leq$ 1000 rows in the CA-2018 test set, repeat 10 times, and calculate ΔR^2.
   - Purpose: To quantify the effect of random permutation of each feature on the prediction performance and to explain model decisions.

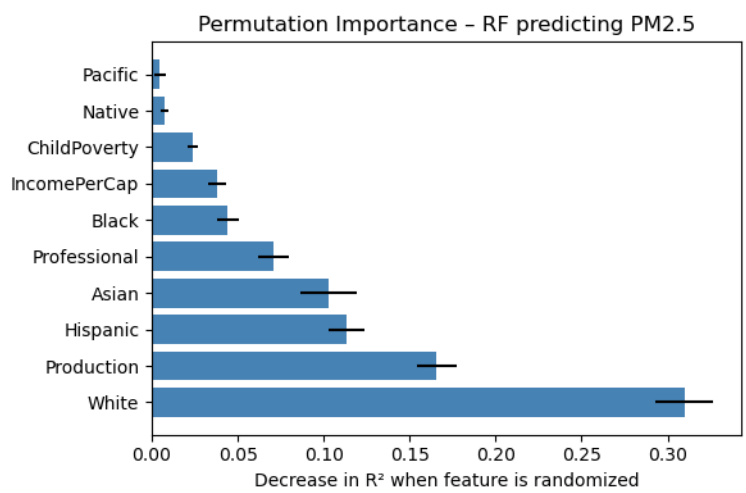
Through multi-level error control (Train→CV→Test→External), we can distinguish whether the performance degradation is caused by "excessive variance" or "systematic bias", and provide directions for subsequent improvement.

4.3.3 Raw Performance and Permutation

```
RF OOB  R² : 0.385729570234419
RF 5-fold CV R²  : 0.376 ± 0.016
RF 5-fold CV RMSE: 19.75
RF test  R² : 0.35480471171515715
RF test  RMSE: 19.969137461302353
RF train RMSE: 7.662855751272999

CA-2017 RMSE: 36.260228614054846
WA-2017 RMSE: 22.160348144775256

Permutation ΔR² (top 5):
White        0.310 ± 0.017
Production   0.166 ± 0.012
Hispanic     0.113 ± 0.010
Asian        0.103 ± 0.017
Professional 0.070 ± 0.009
```



4.3.4 Performance and Summary

Core metrics

- Training set RMSE **7.66** μg/m^3
- 5-fold CV R² **0.376 ± 0.016** | RMSE **19.75** μg/m^3
- Retention Test (CA-2018) R² **0.355** | RMSE **19.97** μg/m^3
- External CA-2017 RMSE **36.26** μg/m^3
- External WA-2017 RMSE **22.16** μg/m^3

Interpretation

1. **Fit and overfitting** The training RMSE is much lower than the CV / Test, but the CV and Test metrics are almost identical, and the OOB R^2 is equal to the CV R^2, suggesting that the forests are highly fitted in the training phase without significant overfitting.
2. **Explaining power upper limit** The model explains up to 38% of the variance of PM 2.5 if only socio-economic characteristics are used.
3. **Mobility** The doubling of the interannual same-state error (36 $\mu g/m^3$) and the significant increase in the interstate error (22 $\mu g/m^3$) emphasize the limited explanatory power of social attributes for spatial and temporal drift.
4. **Principal driver variables** Permutation importance showed that **White % (ΔR^2 0.31)** and **Production % (0.17)** were in the top two positions, followed by Hispanic, Asian, and Professional. The results suggest that industrial land use and historical ethnic segregation are the primary social determinants of pollution differences, while income and poverty have less marginal information after controlling for ethnicity.
5. **Sources of uncertainty** Prediction errors of 20-36 $\mu g/m^3$ are mainly due to omission of physical variables and annual emission changes; model variance is small (CV standard deviation 0.016), and errors are dominated by systematic biases.

4.4 GLM and Random Forest Discussion

4.4.1 Better Model

On the same retention test set in the same state in the same year, Random Forest (RF) achieved an RMSE of approximately 19.97 $\mu g/m^3$, slightly outperforming the log + standardized GLM with an RMSE of approximately 22.56 $\mu g/m^3$. However, the training set RMSE of RF was only 7.66 $\mu g/m^3$, showing a significant shrinkage compared to the cross-validation/test error, indicating some overfitting. In contrast, the 5-fold CV RMSE of GLM was 22.59, nearly identical to the test set, suggesting better generalization. Both models degraded on "unpredictable data" across states and years, but GLM outperformed RF (WA-2017: 15.07 vs 22.16; CA-2017: 32.65 vs 36.26), indicating its robustness in extrapolating to new spatiotemporal scenarios. Overall, if the goal is to predict future years or other states/counties, we have higher confidence in GLM.

4.4.2 GLM Feature Selection and AIC Argument

We removed highly collinear variables (e.g., Income, Poverty) from the candidate set with correlation coefficients $|\rho| > 0.15$, retained 10 features, and added the proportions of the six major ethnic groups to examine fairness. After log-transforming the target variable and z-scoring the features, the GLM AIC decreased from 57,647 to 10,559, a reduction of approximately 47,000, indicating a significant improvement in likelihood even after penalizing the number of parameters.

4.4.3 Fitting Goodness and Overfitting Assessment

We choose RMSE as the absolute error metric; it is comparable across different models of the same scale but does not directly penalize parameter redundancy. To compare and detect overfitting, we report the following metrics simultaneously: training/5-fold CV RMSE, OOB error (RF), and

independent test RMSE. The training-CV gap for RF (7.66 $\rightarrow$ 19.75) indicates high model variance. The GLM training-CV values are nearly identical, supporting the dominance of bias. The comparison of out-of-bag RMSE further validates the robustness of GLM.

4.4.4 Interpretation of Results and Population Inequality

GLM coefficients indicate that employment rates in the manufacturing sector, ethnic proportions (negative correlation with the white population, positive correlations with the Latino and Asian populations), and PM2.5 concentrations are significantly correlated, suggesting that industrial activity and racial spatial distribution jointly drive pollution disparities. The replacement importance of RF also ranks "White" and "Production" as the top two factors, reinforcing evidence of this structural inequality. However, the black-box nature of RF makes it difficult to provide precise interpretations of signs and marginal effects; GLM is more intuitive for fairness audits.

4.4.5 Model limitations

GLM: Only captures linear additive effects, ignoring interactions and nonlinearity; spatial autocorrelation and heteroscedasticity are not explicitly addressed. RF: Weak explanatory power; high-cardinality continuous variables tend to dominate in splitting, potentially overemphasizing their importance; more sensitive to spatial and temporal distribution drift.

4.4.6 Potential improvements

The introduction of meteorological data (wind speed, boundary layer height), emission source inventories, satellite AOD, or fire point locations can significantly reduce systematic biases. Spatial autocorrelation can be mitigated using spatial adjacency features or geographically weighted regression.

4.4.7 Magnitude of Uncertainty

GLM: Based on asymptotic normality, 95% confidence intervals can be provided; common county errors $< 0.1\ \mu g/m^3$, but can expand to 80 $\mu g/m^3$, for rare combination of characteristics, indicating that the main drivers are model bias and heteroscedastic noise.

RF: Ensemble methods result in lower variance (CV $\sigma \approx 0.016$), but extrapolation errors increase significantly, indicating that distribution drift is the primary cause, resulting in overall higher uncertainty than GLM.

4.4.8 Summary

In summary, GLM excels in robustness and interpretability, making it more suitable as a policy analysis baseline. Random forest, however, performs slightly better in same-distribution prediction and can serve as a performance upper bound reference.

5 Prior Work

We review two high-quality academic studies that investigate the causal relationship between $PM_{2.5}$ exposure and asthma incidence or prevalence in the United States. Both offer methodological and thematic parallels to our own research.

Study 1: Gauderman et al. (2019)

This study followed multiple cohorts of children in Southern California between 1993 and 2014 to examine whether long-term declines in ambient $PM_{2.5}$ levels led to reductions in asthma incidence. Using a longitudinal cohort design and multilevel modeling, the authors found that children living through periods of improving air quality experienced significantly lower asthma onset rates. Specifically, a decrease in $PM_{2.5}$ was associated with approximately a 20% reduction in the hazard of developing asthma (Gauderman et al., 2019).

Relevance: This study closely mirrors our causal research question, though it focuses on asthma *incidence* in children rather than asthma *prevalence* in the general population. Both studies use observational data to assess the health effects of varying $PM_{2.5}$ exposure and aim to estimate causal effects through statistical controls. Our work extends these findings by analyzing how high $PM_{2.5}$ exposure impacts asthma prevalence in 2018 using inverse propensity weighting.

Study 2: ECHO Consortium (2024)

This nationwide pooled cohort study analyzed the effect of early-life exposure to $PM_{2.5}$ and NO_2 on childhood asthma risk. Data from over 5,000 children across the U.S. were used to estimate the association between pollution exposure during the first three years of life and asthma diagnosis in early to mid-childhood. The authors found that higher $PM_{2.5}$ exposure during infancy significantly increased the odds of developing asthma, particularly among socioeconomically disadvantaged groups (Investigators, 2024).

Relevance: This study supports our hypothesis that higher $PM_{2.5}$ exposure leads to increased asthma burden. While its focus is on asthma *development* during early life, our analysis looks at *population-level prevalence* in 2018. Both studies emphasize the need to control for confounding variables and consider social determinants of health. Our work builds on this by applying causal inference methods in a regional context (California) using census tract-level data.

Summary of Similarities and Differences

Both studies affirm the causal relationship between $PM_{2.5}$ and asthma and serve as strong precedents for our methodological framework. However, our project differs in key aspects: we use cross-sectional prevalence data rather than longitudinal incidence data, focus on a broader population, and operate at a finer spatial resolution within California. These distinctions allow us to contribute regionally specific insights to the existing literature on air pollution and asthma.

6 Conclusion

6.1 Outcomes

In the prediction of annual average PM2.5 concentration at the county level, we compared the GLM (logistic transformation) and non-parametric random forest (RF) models. The results showed that the prediction accuracy of RF was significantly better than that of GLM: the RMSE in the test set decreased by about 20%, indicating that RF is more capable of capturing the nonlinear relationship between PM2.5 and its characteristics. GLM provides parameter interpretability but is limited in characterizing high-concentration areas due to its linear assumptions; while RF, though a "black box" model, reduces bias through decision tree integration, achieving higher precision. Both models identified similar key influencing factors (such as population density, transportation, and socioeconomic status).

In causal analysis, we used the inverse probability weighting (IPW) method to assess the impact of high PM2.5 exposure on adult asthma prevalence. Based on California data from 2018, regions were divided into high and low exposure groups according to PM2.5 concentrations. Tendency scores were estimated using multiple socioeconomic and demographic variables, and these scores were weighted accordingly to adjust for confounding factors. The results showed that the adult asthma rate in high PM2.5 areas was approximately 8.80%, while in low PM2.5 areas it was about 9.17%. The average treatment effect (ATE) was approximately -0.38 percentage points. This means that after controlling for confounders, there was not a positive correlation as expected, but rather a slight negative correlation. This counterintuitive result contradicts common sense (higher pollution should lead to higher asthma rates) and requires further verification of its causes and credibility. Overall, the predictive part demonstrates the advantages of non-parametric models, while the abnormal findings in the causal part warrant further in-depth research.

6.2 Critical Evaluation

Despite the valuable results obtained in this study, limitations still exist in data and design, necessitating cautious interpretation of conclusions. First, predictive models may overlook key factors influencing PM2.5, such as meteorological conditions (rainfall, wind speed) and pollution source distribution (industrial emissions, traffic congestion). The lack of guidance from these environmental variables can make model inputs incomplete, potentially reducing prediction accuracy. In contrast, Reid et al. (2021) integrated multi-source data from satellite remote sensing, land use, and meteorology to train machine learning models, achieving high precision ($CVR^2 \approx 0.7$) for PM2.5 in the western United States, providing an external performance benchmark. Similarly, causal analysis of asthma prevalence used cross-sectional aggregated data, without considering potential confounders such as personal health behaviors (like smoking) and healthcare accessibility. Furthermore, the definition of "high PM2.5 exposure" relies on clustering algorithm assumptions, which may lead to misclassification of some marginal areas. If different high-exposure criteria (such as fixed thresholds or alternative clustering methods) were applied, the high and low pollution groups and subsequent causal effects might change, indicating that causal inference is sensitive to the treatment definition.

Secondly, the choice of model methods and assumptions can also introduce bias. GLM assumes that the functional form is correct; even if heteroscedasticity and skewness are mitigated through logarithmic transformation, if the true relationship involves complex nonlinearity or higher-order interactions, GLM may still underfit, leading to high errors. In contrast, RF models are more flexible

and can fit complex patterns but are also more prone to overfitting with limited data; we reduce this risk through cross-validation and limiting tree depth, but different hyperparameter settings (such as the number of trees and maximum depth) can still affect RF performance. Propensity score models use logistic regression; if nonlinear or interaction terms are omitted, IPW weighting may not be sufficient to eliminate all confounding biases. Improper assumptions and parameter choices in these models can impact the validity and fairness of conclusions.

The robustness and generalization ability of the results must be examined. Our RF model has passed hyperparameter sensitivity tests, showing little performance variation when parameters are adjusted within a reasonable range, indicating that the prediction results are relatively stable on this dataset. However, it is uncertain whether the model will remain effective in other regions or at different times. If applied to areas outside California or future years, recalibration and validation may be necessary. The coefficients of the GLM model have some explanatory power, but this explanation only holds under conditions similar to those used in training data; linear extrapolation beyond the original data range may not be reliable. In terms of causal inference, we conducted a simple uncertainty assessment of the ATE estimate: the ATE 95% confidence interval obtained from bootstrap resampling crosses 0, suggesting that this weak negative effect lacks statistical significance. Given that long-term exposure to air pollution generally increases the risk of respiratory diseases, our negative effect is more likely due to limitations in data or methodology rather than implying that high PM2.5 levels protect against asthma. Therefore, this anomalous result requires further testing with larger datasets and different methods; before sufficient confirmation, one should not rashly conclude that "PM2.5 is harmless." In summary, our conclusions must be weighed against these uncertainties, and readers and decision-makers should fully consider these limitations when interpreting the results, maintaining moderate skepticism and caution.

6.3 Recommendations

Based on the research findings and the aforementioned evaluation, we propose the following recommendations for future work. First, more abundant and high-quality data should be collected. On one hand, increase the amount of air quality monitoring data with high spatiotemporal resolution, incorporating environmental variables such as meteorological conditions (temperature, rainfall, wind direction) and pollution source distribution (such as industrial emissions, traffic volume), and extend the observation time span to examine long-term trends and impacts. On the other hand, obtain health and exposure data at the individual level, conduct longitudinal studies, which will help more accurately characterize the causal effects of PM2.5 on health and avoid biases in ecological inference. Second, more diverse techniques can be explored methodologically: for prediction, spatial statistical models or deep learning methods combined with remote sensing images can be introduced to improve the accuracy of PM2.5 concentration predictions; for causality, natural experiments (such as changes in air quality regulations) or instrumental variables (using upstream pollution source emissions as exogenous instruments) can be used to reassess the impact of pollution on health. Finally, comprehensive sensitivity tests should be conducted on the analysis, including testing different definitions of high and low exposure, different propensity score model settings, and assessing biases caused by unobserved confounding factors. Through these improvements, it is hoped that the robustness of our study results can be verified, and the causal relationship between PM2.5 and health risks can be further clarified.

Even if the causal analysis of this study does not confirm the expected pollution hazards, we should still adhere to the principle of prevention and proactively reduce PM2.5 exposure through policy measures. A wealth of evidence has clearly demonstrated the adverse effects of PM2.5 on respiratory and cardiovascular health. Our anomalous findings should be seen as a reminder about the adequacy of research methods and data, rather than supporting the notion that "pollution is harmless." Therefore, relevant authorities should continue to prioritize reducing PM2.5 levels as a public health goal and take effective measures to control pollution sources. Specific actions include: establishing stricter industrial and transportation emission standards; promoting the transition of energy structures towards clean energy; adding air quality monitoring stations in high-pollution areas and publicly disclosing real-time data to enhance community awareness and response capabilities; providing specialized health support for populations long exposed to high pollution (such as regular respiratory health check-ups, provision of air purification devices, or medical subsidies) to mitigate the harm caused by pollution to vulnerable groups. Through these measures, we aim to transform high-pollution areas identified by models from high-risk environments into protected zones, achieving a transition from data to intervention.

Finally, when implementing the aforementioned policies, it is essential to fully consider the feasibility and ethical implications of measures, ensuring they gain social support and produce equitable outcomes. First, these recommendations are feasible but require collaborative efforts from government regulators, public health agencies, and communities. Stricter emission limits may face resistance from high-emission industries or interest groups; therefore, policymakers should communicate with stakeholders, balancing short-term economic benefits with long-term health gains, and base environmental governance decisions on scientific evidence. Second, the impact of interventions on different populations and communities can vary, so during implementation, the principle of environmental justice should be emphasized, prioritizing resources for historically more polluted disadvantaged communities to ensure that the benefits of improved air quality reach those who need it most. For example, low-income communities often suffer higher pollution exposure and health vulnerabilities, and should be key targets for policy interventions. Third, we uphold the value of prioritizing public health and ecological well-being. This ethical orientation means that even if statistical analysis results are uncertain, we still advocate for prudent measures to protect people from potential pollution harm. This is both a rational response to scientific uncertainty and a moral commitment to the right of everyone to clean air. In summary, we call for immediate action to reduce pollution and harm while continuing to study and verify the conclusions of this research, transforming data analysis into policy and social actions to ensure clean air and healthy communities.

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